

Introduction to the Classification of Manifolds

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1 Introduction

Showing two manifolds are *not* diffeomorphic is relatively easy, say via homotopy/homology groups. Showing they *are* diffeomorphic is comparatively hard as on the face of it one needs to construct an explicit diffeomorphism between them. In low dimensions, we have shortcuts. The only one-dimensional manifolds (without boundary) are $\mathbb{R}$ and S^1 (See [15]). In dimension 2, orientable surfaces are classified by the Euler characteristic χ . What is remarkable is that progress towards similar classifications in dimensions at least 5 can seem easier than in dimensions 3 and 4. This is essentially because we have more "room to maneuver" in high dimensions. As we will see, this essentially comes down to being able to use the so-called Whitney Trick.

The main tool we will use is the h -cobordism theorem, which roughly says that to show two manifolds are diffeomorphic, instead of explicitly finding a diffeomorphism between them, it is sufficient to find a suitably well-behaved *manifold* which has the disjoint union of the two manifolds as its boundary.

Theorem 1.1. (*Smale 1962, see [21]*)

Suppose W is a simply-connected compact manifold of dimension at least 6, such that $\partial W = \partial_0 W \amalg \partial_1 W$, and the inclusions $\partial_i W \hookrightarrow W$ are homotopy equivalences. Then W is diffeomorphic to $\partial_0 W \times [0, 1]$ relative to $\partial_0 W = \partial_0 W \times \{0\}$. In particular, $\partial_0 W \cong \partial_1 W$.

Note that the requirement for the boundary components to include via homotopy equivalences is in some sense the weakest one can hope for: It is trivially true for a cylinder so for there to be any hope that our cobordism is trivial this is really necessary. The point of the theorem is that it is also sufficient.

The structure of this work is as follows: In Section 2 we follow [13] to develop the theory of handlebody decompositions of cobordisms and use it to prove the h -cobordism theorem, which we then apply to prove the Poincaré conjecture in dimensions at least 5. Section 3 follows the proof of a theorem of Kreck from [9] which gives sufficient conditions for two $2n$ -dimensional manifolds to be stably diffeomorphic (that is, diffeomorphic after possibly first connect-summing with copies of $S^n \times S^n$). Then, Section 4 follows the proof of another theorem of Kreck's from [10] which shows when we can destabilize such a diffeomorphism. Finally, in Section 5 we apply these tools to build an exotic 8-sphere Σ such that $\mathbb{H}\mathbb{P}^2 \# \Sigma$ is diffeomorphic to $\mathbb{H}\mathbb{P}^2$, thus proving a special case of the main theorem in [12]. We in general have specialized to the simply-connected case.

2 The h -cobordism theorem

In this section we develop the theory of Handlebody Decompositions and use it to prove the h -cobordism theorem. We follow [13] and specialize to the simply-connected case. The primary object of study is a compact *cobordism* between two oriented manifolds M and N . This is a compact oriented manifold W such that $\partial W = \partial_0 W \amalg \partial_1 W$ equipped with orientation-preserving diffeomorphisms $f_0 : M \rightarrow \partial_0 W$ and $f_1 : N \rightarrow \partial_1 W$. An h -cobordism is a compact cobordism (W, M, N) such that the inclusions $M \hookrightarrow W, N \hookrightarrow W$ are homotopy equivalences. In this work we will only ever consider cobordisms between simply connected manifolds, and note that these are always orientable so they fit our context.

We briefly record the following fact which is elementary but worth mentioning as it is used often in this work.

Lemma 2.1. *A compact cobordism $(W, \partial_0 W, \partial_1 W)$ is an h -cobordism if and only if the inclusions $\partial_{0,1} W \hookrightarrow W$ induce isomorphisms on π_1 , and $H_*(W, \partial_{0,1} W) \cong 0$.*

Proof. This follows from the long exact sequence on relative homology as well as Whitehead's Theorem. □

2.1 Handlebodies and Handlebody Decompositions

The importance of cell structures in algebraic topology is that cells give precise control over algebraic information such as π_n and H_n . For instance, one can easily kill an element of π_r of some space by attaching a D^{r+1} along its boundary. We can't, however, perform exactly this operation on a manifold if we wish for the result to remain a manifold. A very clear problem is that if for instance $\dim M > 2$ and we want to kill an element of $\pi_1(M)$, attaching a 2-cell is just the wrong dimension to be working with. The idea to solve this issue is to first thicken the cell you'd like to be attaching, to get a *handle*.

Definition 2.1. *The n -dimensional q -handle is $D^q \times D^{n-q}$. Its core is $D^q \times \{0\}$, and its transverse sphere is $\{0\} \times S^{n-q-1}$.*

We will be gluing these along the $S^{q-1} \times D^{n-q}$ portion of their boundary. If $\phi^q : S^{q-1} \times D^{n-q} \rightarrow \partial M$ is an embedding, we write $M + (\phi^q) := M \cup_{\phi^q} D^q \times D^{n-q}$. It turns out that handlebodies are universal enough that every cobordism can be decomposed into them from a trivial bordism (a cylinder), where importantly we glue entirely onto just one component of the boundary. That is, given a compact manifold W such that $\partial W = \partial_0 W \amalg \partial_1 W$, there is a finite sequence of compact manifolds $W_0, W_1, \dots, W_r$ defined by

$$W_0 = \partial_0 W \times [0, 1]$$

$$W_i = W_{i-1} + (\phi_i^{q_i})$$

where we set $\partial_0 W_0 = \partial_0 W \times \{0\}$ and then always have $\text{im}(\phi_i^{q_i}) \subseteq \partial_1 W_{i-1}$ (thus leaving $\partial_0 W_i := \partial_0 W \times \{0\}$ untouched at every step), such that

$$W \cong W_r = \partial_0 W \times [0, 1] + \sum_{i=1}^r (\phi_i^{q_i})$$

where the diffeomorphism is relative to $\partial_0 W = \partial_0 W \times \{0\}$. (See chapter 6 of [6])

The h-cobordism theorem requires reducing a cobordism to a trivial one, so to make progress we'd like to understand when and how we can remove these added handlebodies, without changing the diffeomorphism class of the manifold relative to $\partial_0 W$.

We are going to record here some lemmata about handle manipulations which we will mostly not prove as the proofs are quite technical, and remain virtually unchanged in the simply-connected case. Proofs are in [13]. We *will* prove the Normal Form Lemma for cobordisms as Lück's hypotheses in [13] can be slightly weakened, and this will be key to proving the stabilization theorem later on.

Lemma 2.2. *(Isotopy Lemma)*

Let W be a compact n -dimensional manifold such that $\partial W = \partial_0 W \amalg \partial_1 W$, and suppose

$$\phi^q, \psi^q : S^{q-1} \times D^{n-q} \rightarrow \partial_1 W$$

are isotopic embeddings. Then, $W + (\phi^q)$ and $W + (\psi^q)$ are diffeomorphic relative to $\partial_0 W$.

This is a very intuitive result. The next result states that if two cobordisms W and W' are diffeomorphic via a map f also restricting to a diffeomorphism $f|_{\partial_0 W} : \partial_0 W \rightarrow \partial_0 W'$, then attaching a handlebody to $\partial_1 W$ essentially commutes with the diffeomorphism.

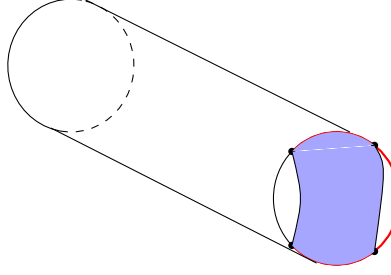
Lemma 2.3. (*Diffeomorphism Lemma*)

Suppose W and W' are as above, with $F : W \rightarrow W'$ a diffeomorphism such that the restriction $F|_{\partial_0 W} : \partial_0 W \rightarrow \partial_0 W'$ is also a diffeomorphism. Given an embedding $\phi^q : S^{q-1} \times D^{n-q} \rightarrow \partial_1 W$, there is an embedding $\bar{\phi}^q : S^{q-1} \times D^{n-q} \rightarrow \partial_1 W'$ and a diffeomorphism $F' : W + (\phi^q) \rightarrow W' + (\bar{\phi}^q)$ inducing $F|_{\partial_0 W}$ on $\partial_0 W$.

We will now give a typical example of when we can cancel handles. We will see that we will in fact be cancelling *two* handles of consecutive indices q and $q + 1$. In general, let $(W, \partial_0 W, \partial_1 W)$ be an n -dimensional cobordism, and let $0 \leq q \leq n - 1$. We will illustrate the construction by considering an example with $n = 2, q = 1$, and $W \cong S^1 \times [0, 1]$. In general, consider an embedding

$$\mu : D^{n-1} \cong S^{q-1} \times D^{n-q} \cup_{S^{q-1} \times S_+^{n-1-q}} D^q \times S_+^{n-1-q} \rightarrow \partial_1 W$$

In our example, this is an embedding $D^1 \hookrightarrow S^1$, which will be a $3/4$ arc for illustration. Let ϕ^q be the restriction of μ to $S^{q-1} \times D^{n-q}$. We will then consider how to add a $q + 1$ handle to $W + (\phi^q)$, which in our example looks like:



We will build our $q + 1$ handle ψ^{q+1} out of two components $\phi_+^{q+1} : S_+^q \times S_+^{n-q-1} \rightarrow \partial_1(W + (\phi^q))$ and $\phi_-^{q+1} : S_-^q \times S_+^{n-q-1} \rightarrow \partial_1(W + (\phi^q))$. ϕ_+^{q+1} is defined simply by the embedding

$$S_+^q \times S_+^{n-q-1} = D^q \times S_+^{n-q-1} \subset D^q \times S^{n-q-1}$$

and this last space is of course the boundary of the q -handle we've just glued on. In our picture we can consider thus that we are gluing half of the boundary of our 2-handle (Since a 2-handle in dimension 2 is just $D^2 \times \{0\}$, this boundary is of course just S^1) to the rightmost component of the boundary of our 1-handle. ϕ_-^{q+1} is then defined by identifying $S_-^q \times S_+^{n-q-1} = D^q \times S_+^{n-q-1}$ and using the embedding μ restricted to this part of its source. In our example thus the second half of our S^1 is glued onto the red arc that has not been touched yet. The effect of gluing ψ^{q+1} on in our illustration is then clearly just filling in the rightmost gap with a D^2 , and it's then easy to see that $S^1 \times [0, 1] + (\phi^q) + (\psi^{q+1}) \cong S^1 \times [0, 1] \text{ rel } S^1 \times \{0\}$.

Indeed, in the general case the effect of adding both handles is just doing the boundary connected sum of W and D^n along the embedding μ of $D^{n-1} \subset \partial D^n$ into $\partial_1 W$. This construction can be generalised to get the following results:

Lemma 2.4. (*Cancellation Lemma*) Suppose $\phi^q : S^{q-1} \times D^{n-q} \rightarrow \partial_1 W$ and $\psi^{q+1} : S^q \times D^{n-q-1} \rightarrow \partial_1(W + (\phi^q))$ are embeddings such that $\psi^{q+1}(S^q \times \{0\})$ meets the transverse sphere of (ϕ^q) transversally and in exactly one point. Then, there is a diffeomorphism $W \cong W + (\phi^q) + (\psi^{q+1})$ relative to $\partial_0 W$.

Lemma 2.5. (*Elimination Lemma*) Suppose $1 \leq q \leq n-3$. Suppose W has a handlebody decomposition with no handles of index less than q , that is

$$W \cong \partial_0 W \times [0, 1] + \sum_{i=1}^{p_q} (\phi_i^q) + \dots + \sum_{i=1}^n (\phi_i^n)$$

Given an integer $1 \leq i_0 \leq p_q$, suppose there is an embedding $\psi^{q+1} : S^q \times D^{n-1-q} \rightarrow \partial_1^\circ W_q$ such that $\psi^{q+1}|_{S^q \times \{0\}}$ is isotopic in $\partial_1 W_q$ to an embedding which meets the transverse sphere of $(\phi_{i_0}^q)$ transversally and in exactly one point, and which is also disjoint from the transverse spheres of all other q -handles. Suppose further that $\psi^{q+1}|_{S^q \times \{0\}}$ is isotopic in $\partial_1 W_{q+1}$ to a trivial embedding. Then, we can cancel the handle $(\phi_{i_0}^q)$ at the cost of adding a $q+2$ handle and changing handles of index greater than q . That is, W is diffeomorphic relative to $\partial_0 W$ to

$$\partial_0 W \times [0, 1] + \sum_{1 \leq i \leq p_q, i \neq i_0} (\phi_i^q) + \sum_{i=1}^{p_{q+1}} (\bar{\phi}_i^{q+1}) + (\psi^{q+2}) + \dots + \sum_{i=1}^{p_n} (\bar{\phi}_i^n)$$

The next lemma is the beginning of properly simplifying the form of a handlebody decomposition. It will be used as the base for a much stronger simplification later.

Lemma 2.6. Let $(W, \partial_0 W, \partial_1 W)$ be a compact n -dimensional cobordism for $n \geq 6$. If the inclusion $\partial W_0 \rightarrow W$ is 1-connected, then we can find a diffeomorphism relative $\partial_0 W$

$$W \cong \partial_0 W \times [0, 1] + \sum_{i=1}^{p_2} (\phi_i^2) + \dots + \sum_{i=1}^{p_n} (\phi_i^n)$$

That is, W has a handlebody decomposition with no 1-handles.

We will want to understand what homological information arises from a handlebody decomposition and how we may modify it. This is in some sense quite simple, as it is clear that gluing a q -handle is homotopically the same as gluing on a q -cell. This is true via the obvious equivalences $D^q \times D^{n-q} \simeq D^q \times \{0\}$ and $S^{q-1} \times D^{n-q} \simeq S^{q-1} \times \{0\}$. Everything turns out to work nicely and indeed we get that

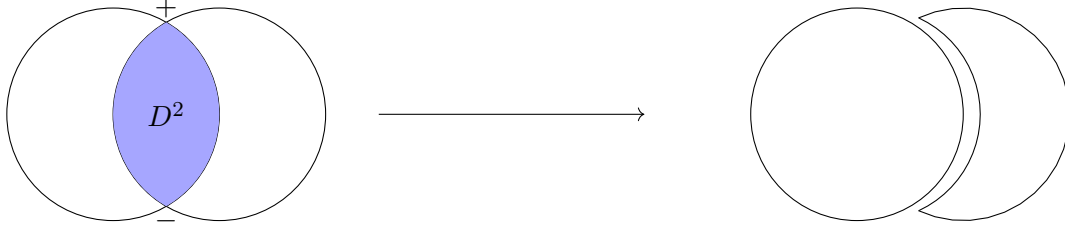
$$C_q(W, \partial_0 W) = \langle \{[\phi_i^q] : i = 1, \dots, p_q\} \rangle$$

Where $[\phi_i^q]$ is the image of the characteristic map of the pair $(D^q \times D^{n-q}, S^{q-1} \times D^{n-q})$ (See details in [13] Section 1.2). The key in the following work is understanding when one can pass between gluing information and homological information. We will prove a lemma which says essentially that one can separate (via isotopy) the image of a sphere in $\partial_1 W_q$ from all but one handle (ϕ_i^q) if and only if this image has relative homology class $[\phi_i^q]$.

This will be the first invocation of our high-dimensional context, and the key point is that in high dimensions the *Whitney trick* holds. The following is really the key reason why high-dimensional topology can often be more tractable than lower dimensions.

Lemma 2.7. (*The Whitney Trick*) (See [14]) Suppose M and N are two submanifolds of dimensions r and s intersecting transversally in some simply-connected manifold V without boundary, of dimension $r+s$. If $p, q \in M \cap N$ have opposite intersection number (determined by whether or not the orientations on $T_{p,q}M$ and $T_{p,q}N$ match), then there is an isotopy $H : V \times [0, 1] \rightarrow V$ with $H(\cdot, t) \equiv \text{id}$ near $M \cap N - \{p, q\}$, and with $H(M, 1) \cap N = M \cap N - \{p, q\}$.

The idea is to find an embedded 2-disc which allows the two manifolds to be isotoped off of each other. We illustrate the model in the plane $\mathbb{R}^2$.



The point is that given the dimensionality assumptions, such a 2-disc can always be found. The move performed is then just a stabilization of the above by cartesian products with $\mathbb{R}^N$ for some N .

Lemma 2.8. (*Homology Lemma*) Suppose $n \geq 6$, $2 \leq q \leq n - 3$, and that $\pi_1(W) \cong \{1\}$. Fix some $1 \leq i_0 \leq p_q$. An embedding $f : S^q \rightarrow \partial_1 W_q$ is isotopic to an embedding $g : S^q \rightarrow \partial_1 W_q$ such that g meets the transverse sphere of $(\phi_{i_0}^q)$ transversally and in exactly one point and is disjoint from the transverse spheres of all other q -handles if and only if $[f] = \pm[\phi_{i_0}^q] \in H(W_q, W_{q-1})$.

Proof. The forward direction is straightforward. We can simply isotope f so that the upper hemisphere of S^q lands exactly on $D^q \times \{x\} \subset D^q \times S^{n-q-1}$ for some x on the transverse sphere. The homology result is then clear.

For the reverse direction, note first that by a general position argument we may assume that $\text{im} f$ intersects the transverse spheres of $\{(\phi_i^q)\}$ transversally. Since the dimension of the domain of f is q , and that of the transverse spheres is $n - q - 1$, since $q + (n - q - 1) = n - 1 = \dim \partial_1 W_q$, $\text{im} f$ intersects the transverse sphere of (ϕ_i^q) at finitely many points, say $x_{i,1}, \dots, x_{i,r_i}$. Define $\epsilon_{i,j}$ to be 1 if the orientations of $T_{f^{-1}(x_{i,j})} S^q$ and the tangent space of the transverse sphere of (ϕ_i^q) at $x_{i,j}$ match. We see that

$$[f] = \sum_{i=1}^{p_q} \sum_{j=1}^{r_i} \epsilon_{i,j} [\phi_i^q]$$

Now, by assumption we thus have

$$\pm[\phi_{i_0}^q] = \sum_{i=1}^{p_q} \sum_{j=1}^{r_i} \epsilon_{i,j} [\phi_i^q]$$

Since $\{[\phi_i^q]\}$ is a basis, if $\sum_{i=1}^{p_q} r_i > 1$ then there are i, j_1, j_2 such that $\epsilon_{i,j_1} = -\epsilon_{i,j_2}$. We may then use the Whitney Trick to isotope f to miss these two intersections. Iterating this gives the result. $\square$

The next lemma shows that we can use a handlebody decomposition to edit the homological data of an embedded sphere via isotopy. We are essentially "sliding" q -spheres around in the one-higher dimensional skeleton to induce a change in homological data via catching $q + 1$ -handles and bringing them down via the boundary map.

Lemma 2.9. (*Modification Lemma*) Suppose $\pi_1(W) \cong \{1\}$. Let $f : S^q \rightarrow \partial_1^\circ W_q$ be an embedding and let $\{x_i\}_{i=1}^{p_{q+1}}$ be any integers. Then there is an embedding $g : S^q \rightarrow \partial_1^\circ W_q$ such that f is isotopic to g in $\partial_1 W_{q+1}$ and, in $C_q(W)$,

$$[g] = [f] + \sum_{i=1}^{p_{q+1}} x_i \cdot d_{q+1}[\phi_i^{q+1}]$$

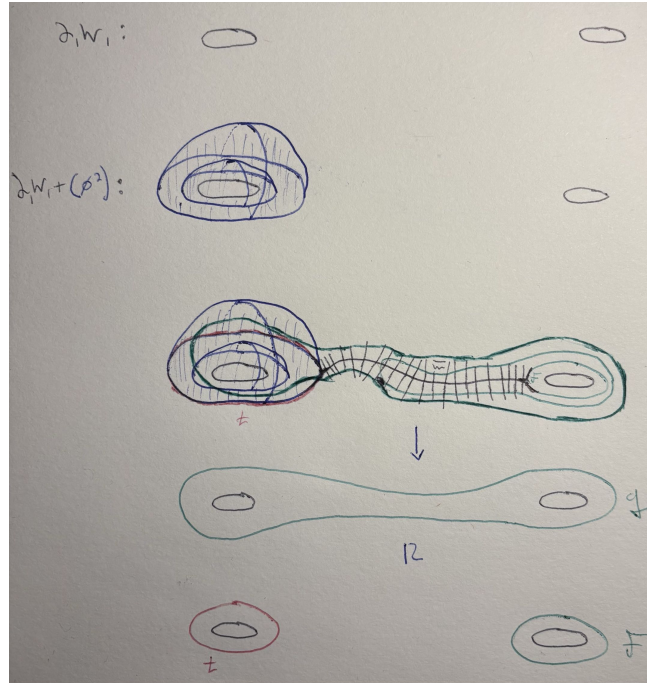
Proof. We obviously only need to show that given any $q+1$ -handle (ϕ_j^{q+1}) and $\epsilon \in \{\pm 1\}$, we can find a suitable $g_\epsilon : S^q \rightarrow \partial_1^\circ W_q$ with

$$[g_\epsilon] = [f] + \epsilon \cdot d_{q+1}[\phi_j^{q+1}]$$

We will accompany the explanation by an example, with $\dim W = 2$, $q = 1$, and $\partial_1 W_1$ is D^2 minus two interior discs. We will be gluing on a 2-handle. One can see $D^2 \times D^1$ as an annulus with a disc glued onto each concentric circle along the identity $\partial D^2 \rightarrow S^1$. Adding a 2-handle is thus essentially gluing a beanie on along its rim.

In general, take some embedding $t : S^q \xrightarrow{\text{id} \times \{z\}} S^q \times \{z\} \subset S^q \times S^{n-q-2} \subset \partial(\phi_j^{q+1}) \cap \partial_1 W_q$. It is of course isotopic to trivial embedding in ∂W_{q+1} as it bounds $D^{q+1} \times \{z\}$, there. This map t of course has homology $d_{q+1}([\phi_j^{q+1}])$, and this is how we will add this homology to $[f]$. We thus want to somehow "connect" these two maps to form just one $S^q \hookrightarrow \partial_1 W_q$. We may connect some two points in the images of t and f by an embedding $w : [0, 1] \hookrightarrow \partial_1^\circ W_q$ which is otherwise disjoint from these images (connect the images by *some* path and then take a sub-path with no extraneous intersections), and then thicken this embedding to an $\bar{w} : [0, 1] \times D^q \hookrightarrow \partial_1^\circ W_q$ which intersects the images of t and f in small discs D^q and which is disjoint from them elsewhere.

We can now form the connect-sum of the maps t and f , denoted $g := t \#_w f$. The idea is to write $S^q = D^q \cup S^{q-1} \times [0, 1] \cup D^q$ (a cylinder with the holes capped). We then say that g restricted to the first and second D^q is t and f respectively, and g maps the cylindrical part along the small cylinder that is the "boundary" $\text{im}(w|_{[0,1] \times S^{q-1}})$ of the thickening. As t is isotopic to a trivial embedding in $\partial_1 W_{q+1}$, g is isotopic to f there. However, the image of g is of course homotopic to the disjoint union of the images of f and g in $\partial_1 W_q$ so as chains $[g] = [f] + d_{q+1}(\phi_j^{q+1})$. To get the same equation with a $-$ instead of a $+$, we just pre-compose g by some self-diffeomorphism of S^q of degree -1 .



□

We now have enough to show that every cobordism with sufficiently simple homological information can be reduced to only having handles in two consecutive dimensions. This reduction will make the h -cobordism theorem tractable. Note that our hypothesis is slightly weaker than the one in [13].

Lemma 2.10. (*Normal form for handlebody decompositions*)

Let $(W, \partial_0 W, \partial_1 W)$ be a compact cobordism of dimension $n \geq 6$ such that $H_i(W, \partial_0 W) \cong H_i(W, \partial_1 W) \cong 0$ for $0 \leq i \leq k$. Let q be an integer with $\max(2, n - k - 2) \leq q \leq \min(k + 1, n - 3)$. Then there is a diffeomorphism relative $\partial_0 W$

$$W \cong \partial_0 W \times [0, 1] + \sum_{i=1}^{p_q} (\phi_i^q) + \sum_{i=1}^{p_{q+1}} (\phi_i^{q+1})$$

Proof. We would first like to show that we do not need any handles up to level q . This has been shown in Lemma 2.6 for $q = 2$. To show this is true for $q + 1$ we will show that we can decrease the number of q -handles by adding a $q + 1$ handle and a $q + 2$ handle such that on the one hand these added handles cancel each other out, and on the other hand the $q + 1$ handle cancels with some given q -handle.

Let $\bar{\psi}^{q+1} : S^q \times D^{n-1-q} \rightarrow \partial_q^\circ W_q$ be some nullhomotopic embedding (say within some coordinate chart), and write $\alpha := \bar{\psi}^{q+1}|_{S^q \times \{0\}}$. As we've seen, $C_k(W, \partial_0 W)$ is generated by $\{[\phi_i^k] : i = 1, \dots, p_k\}$. Thus, $C_{q-1}(W, \partial_0 W) \cong 0$, so that since by assumption $H_q(W, \partial_0 W) \cong 0$, the differential d_{q+1} has $\text{im } d_{q+1} = \ker d_q = C_q(W, \partial_0 W)$. There are thus integers $\{s_i\}$ such that

$$[\phi_1^q] = \sum_{i=1}^{p_{q+1}} s_i \cdot d_{q+1}[\phi_i^{q+1}]$$

The Modification Lemma 2.9 then shows that there exists an embedding $\psi^{q+1} : S^q \times D^{n-1-q} \rightarrow \partial_1^\circ W$ such that $\beta := \psi^{q+1}|_{S^q \times \{0\}}$ and α are isotopic in $\partial_1 W_{q+1}$, and such that

$$[\beta] = [\alpha] + \sum_{i=1}^{p_{q+1}} s_i d_{q+1}[\phi_i^{q+1}] = [\phi_1^q]$$

The Homology Lemma 2.8 shows that β is isotopic in $\partial_q W_q$ to an embedding meeting the transverse sphere of (ϕ_1^q) transversally in exactly one point and that is disjoint from all other transverse spheres of handles of index q . Since β is isotopic in $\partial_1 W_{q+1}$ to the trivial embedding $\bar{\psi}^{q+1}$ by construction, the Elimination Lemma 2.5 shows that we can kill (ϕ_1^q) . Thus we can force $W_{-1} = W_{q-1}$.

We now want to build a "dual" handlebody decomposition of W . The details are in [13], but the point is that any cobordism $T = \partial_0 T \times [0, 1] + (\phi^q)$ can also be written as $\partial_1 T \times [0, 1] + (\psi^{n-q})$ for some $(n - q)$ -handle. We can thus at this stage write

$$W = \partial_1 W \times [0, 1] + \sum_{i=1}^{p_n} (\psi_i^0) + \sum_{i=1}^{p_{n-1}} (\psi_i^1) + \dots + \sum_{i=1}^{p_q} (\psi_i^{n-q})$$

Now of course $\max(2, n - k - 2) \leq q \leq \min(n - 3, k + 1)$ trivially implies that $n - q - 1$ is also in this range, so we may apply the first part of the argument to this new decomposition to get our normal form. □

This normal form lemma obviously in particular holds for h -cobordisms, by the long exact sequence on relative homology. Indeed in this case $H_*(W, \partial_0 W) \cong 0$ and so the only non-trivial boundary map

d_{q+1} is an isomorphism. It can thus be represented by some invertible square integer matrix A . The strategy of the proof of the h -cobordism theorem is then to show that certain algebraic operations on A correspond to geometric manipulations on the handlebody decomposition of W , and then to use the fact that we may trivialize A algebraically to show that we may also do so geometrically. We will consider 4 possible operations on A :

1. Adding the k -th row of A multiplied by $n \in \mathbb{Z}$ to the l -th row of A for $k \neq l$.
2. Extending A by direct sum with the matrix $I_1 = (1)$, that is the move $A \rightarrow \begin{pmatrix} A & 0 \\ 0 & 1 \end{pmatrix}$
3. Identifying A as a direct sum with the matrix I_1 and cancelling this factor, that is the move $A = \begin{pmatrix} B & 0 \\ 0 & 1 \end{pmatrix} \rightarrow B$
4. Swapping any two rows of A or any two of its columns.

We will show that for any of these moves our h -cobordism is diffeomorphic relative to $\partial_0 W$ to one whose $(q+1)$ th differential is represented by the new matrix. It then follows from the Smith Normal Form for integer matrices (See [3]) that any invertible square matrix can be made into the identity matrix by these moves, and then applying operation 3 iteratively we see that we can reduce our cobordism to a trivial one.

For the first move, consider the cobordism

$$W' = \partial_0 W \times [0, 1] + \sum_{i=1}^{p_q} (\phi_i^q) + \sum_{i=1, i \neq l}^{p_{q+1}} (\phi_i^{q+1})$$

By the Modification Lemma 2.9 there is an embedding $\bar{\phi}_l^{q+1} : S^q \times D^{n-q-1} \rightarrow \partial_1 W'$ such that $\bar{\phi}_l^{q+1}$ and ϕ_l^{q+1} are isotopic and

$$\left[\bar{\phi}_l^{q+1} |_{S^q \times \{0\}} \right] = \left[\phi_l^{q+1} |_{S^q \times \{0\}} \right] + n \cdot d_{q+1}([\phi_k^{q+1}])$$

By the Isotopy Lemma 2.2, $W' + (\bar{\phi}_l^{q+1}) \cong W$ relative to $\partial_0 W$, and the matrix representing the new d_{q+1} is as required.

For the second move, let $\phi^q : S^{q-1} \times D^{n-q} \rightarrow \partial_1 W$ be some nullhomotopic embedding, and let $\psi^{q+1} : S^q \times D^{n-q-1} \rightarrow \partial_1(W + (\phi^q))$ be an embedding such that $\psi^{q+1}(S^q \times \{0\})$ intersects the transverse sphere of (ϕ^q) transversally and in exactly one point, and is also disjoint from the transverse spheres of all other q -handles. Then, by the Cancellation Lemma Lemma 2.4, W is diffeomorphic to $W + (\phi^q) + (\psi^{q+1})$ rel $\partial_0 W$. We see that $d_{q+1}[\psi^{q+1}] = [\psi^{q+1} |_{S^q \times \{0\}}]$ by just considering W as a cell complex and by basic properties of homology on such spaces. We then see $[\psi^{q+1} |_{S^q \times \{0\}}] = [\phi^q]$ by the Homology Lemma 2.8. This shows that we can realize the second move. The third move is essentially the same argument moving backwards. The fourth move is realized by just swapping the relevant indices in the handlebody decomposition. We have thus proved the h -cobordism theorem.

It is worth noting that by [24], if one follows the proof of the h -cobordism theorem as far as one can when the boundary manifolds have dimension 4, we see that while in general the full theorem is not true, such a setup does imply the two 4-manifolds are *stably* diffeomorphic, that is diffeomorphic after possibly connect-summing with copies of $S^2 \times S^2$.

2.2 The high-dimensional Poincaré Conjecture

In this subsection we apply the h -cobordism theorem to prove:

Theorem 2.1. (*Poincaré Conjecture in dimensions ≥ 5*) Let $n \geq 5$. Suppose Σ is a simply-connected n -manifold and has $H_*(\Sigma) \cong H_*(S^n)$. Then, Σ is homeomorphic to S^n .

Proof. We partly follow both [13] and [14].

Now let $D_0^n, D_1^n \subset \Sigma$ be two disjoint embedded disks. Take $W = \Sigma - (\text{int}(D_0^n) \amalg \text{int}(D_1^n))$. It will turn out that W is an h -cobordism. Indeed, note that $H_i(W, \partial D_0^n) \cong \tilde{H}_i(\Sigma - \text{int}(D_1^n))$. Further, by Lefschetz duality $H_i(\Sigma - \text{int}(D_1^n)) \cong H^{n-i}(\Sigma - \text{int}(D_1^n), \partial D_1^n)$. By excision, this is $H^{n-i}(\Sigma, D_1^n) \cong \tilde{H}^{n-i}(\Sigma)$. Thus $H_*(W, \partial D_0^n) \cong 0$, so by Whitehead's theorem and the long exact sequence on relative homology $\partial D_0^n \hookrightarrow W$ is a homotopy equivalence. The same is clearly true for $\partial D_1^n \hookrightarrow W$.

Thus, by the h -cobordism theorem there is a diffeomorphism F from the trivial cobordism $(\partial D_0^n \times [0, 1], \partial D_0^n \times \{0\}, \partial D_1^n \times \{1\})$ to $(W, \partial D_0^n, \partial D_1^n)$ whose restriction to $\partial D_0^n \times \{0\}$ is the identity, and which induces some diffeomorphism $f_1 : \partial D_0^n \times \{0\} \rightarrow \partial D_1^n$. We can extend f_1 radially to an $\bar{f}_1 : D_0^n \rightarrow D_1^n$ (via $\bar{f}_1(t \cdot x) = t f_1(x)$, $t \in [0, 1], x \in S^{n-1}$). The homeomorphism

$$h : S^n \cong D_0^n \times \{0\} \cup_{i_0} \partial D_0^n \times [0, 1] \cup_{i_1} D_1^n \times \{1\} \rightarrow \Sigma$$

Defined by $h|_{D_0^n \times \{0\}} = \text{id}$, $h|_{\partial D_0^n \times [0, 1]} = F$, and $h|_{D_1^n \times \{1\}} = \bar{f}_1$ then shows $\Sigma \cong S^n$.

In dimension 5 we use the fact from ([14] Theorem 9.2) that $\Sigma = \partial M$ for some contractible manifold M . Let $D \subseteq \text{int}(M)$ be a disc and denote $W := M - \text{int}(D)$. Obviously, $H_*(W, \partial D) \cong \tilde{H}_*(M) \cong 0$. Also, by Lefschetz Duality for triples, $H_*(W, \Sigma) = H^{6-*}(W, \partial D) \cong 0$ by universal coefficients. Thus W is an h -cobordism and so Σ is in fact *diffeomorphic* to S^5 . □

This last result was stronger for dimension 5. In fact, Theorem 9.2 in [14] also says that a homotopy 6-sphere is the boundary of a contractible manifold so the same argument shows that it must be diffeomorphic to S^6 . We'll see in the final section that it is not true in all dimensions that homotopy spheres are diffeomorphic to the standard sphere, and indeed whether it is true in dimension 4 is still open. The question in the topological category was of course famously much harder in dimension 3. See [2] for an exposition in this dimension.

3 Stabilisation by $S^n \times S^n$

We will now use the proof of the h -cobordism theorem (The Normal Form lemma specifically) to find sufficient conditions for when two $2n$ -dimensional manifolds are diffeomorphic, up to first connect-summing by copies of $S^n \times S^n$. This follows [9]. The conditions involved are stable homotopy-theoretic.

We first need a brief discussion on normal B -structures. Recall that a vector bundle $E \rightarrow M$ is stably classified by a homotopy class of maps $M \rightarrow BO$.

Given a fibration $B \rightarrow BO$, a normal B -structure on a manifold M is a lift $\bar{\nu} : M \rightarrow B$ of the stable normal Gauss map $\nu : M \rightarrow BO$ corresponding to the normal bundle of M when embedded in some sufficiently high-dimensional Euclidean space. Such a pair $(M, \bar{\nu}_M)$ is called a B -manifold. A normal B -structure $\bar{\nu} : M \rightarrow B$ is said to be a *normal k -smoothing* if it is a $(k+1)$ -equivalence, i.e. if it induces isomorphisms in π_i for $i \leq k$, and also a surjection $\pi_{k+1}(M) \rightarrow \pi_{k+1}(B)$.

Also, a fibration $B \rightarrow BO$ is said to be *k -universal* if its fibre is connected with vanishing homotopy groups in dimensions $\geq k+1$.

Proposition 3.1. *Let M be a manifold. There is a tower of fibrations $\dots \rightarrow B_{s+1} \rightarrow B_s \rightarrow \dots \rightarrow B_1 \rightarrow BO$ such that for all $k \geq 1$ the fibration $B_k \rightarrow BO$ is k -universal, and there exists a normal k -smoothing $\bar{\nu} : M \rightarrow B_k$. Moreover, any two such B_k are fibre homotopy equivalent.*

(See [10]) This tower of fibrations is the *Postnikov decomposition* of the map $M \xrightarrow{\nu_M} BO$. Thus, the fibre homotopy type of a k -universal fibration $B \rightarrow BO$ admitting a normal k -smoothing of some manifold M is well defined. It is denoted $B_k(M)$.

Two closed B -manifolds M and N are called *B -bordant* if there is a cobordism (W, M, N) equipped with a normal B -structure $\bar{\nu} : W \rightarrow B$ such that $\bar{\nu}_W|_M = \bar{\nu}_M$ and $\bar{\nu}_W|_N = -\bar{\nu}_N$.

We can now state the main result of this section.

Theorem 3.1. *Suppose M and N are two closed $2q$ -dimensional manifolds with the same Euler characteristic, such that $B_{q-1}(M) = B_{q-1}(N)$. Suppose also that the $(q-1)$ -smoothings $\bar{\nu}_M$ and $\bar{\nu}_N$ are $B_{q-1}(M)$ -bordant. Then $M \#_r(S^q \times S^q)$ and $N \#_r(S^q \times S^q)$ are diffeomorphic for some r .*

The idea will be to do surgery on such a bordism to make its relative homology simple enough that the Normal Form lemma applies, and we'll do this by forcing the B -structure to be a $(q-1)$ -smoothing. We must thus discuss some surgery theory.

The main idea is that we can kill elements of π_r by attaching a handle, as long as they can be represented by an embedded S^r with trivial normal bundle. The fundamental observation is that $\partial(S^r \times D^{m-r}) = \partial(D^{r+1} \times S^{m-r-1})$. Thus we can delete a tubular neighborhood of an embedded S^r with trivial normal bundle, and glue on a $D^{r+1} \times S^{m-r-1}$ in its place. This will kill the element of π_r represented by the embedded S^r . It turns out that, *in the range* $r < q$, elements in the kernel of $\pi_r(M) \rightarrow \pi_r(B)$ can always be represented by an embedding with trivial normal bundle, so we can always force a normal structure to be a $(q-1)$ -smoothing.

To perform such a surgery, we will actually construct a B -bordism W between our manifold M and the result of the surgery M' . W will be called the *trace* of the surgery on M . Explicitly, W is $M \times [0, 1] + (f^{q+1})$. This is *a priori* just a normal cobordism, but we'll soon show that we can extend the B -structure on M to make it a B -bordism.

Lemma 3.1. *(Effect of surgery on π_r) Suppose M is connected and $f : S^r \times D^{m-r} \rightarrow M$ is an embedding, with $2r < m-1$. If M' is the result of a surgery on M , then $\pi_r(M') \cong \pi_r(M)/\langle\langle [f] \rangle\rangle$.*

Proof. Since $W \simeq D^{r+1} \cup_{f|_{S^r \times \{0\}}} M$, we're homotopically just gluing on an $(r+1)$ -cell and then it's standard that $M \hookrightarrow W$ is r -connected with the kernel of the induced map on π_r equal to $\langle\langle [f] \rangle\rangle$. Dually, $W \simeq D^{m-r} \cup_{\{0\} \times S^{m-r-1}} M'$ and so $M' \hookrightarrow W$ is an $(m-r-1)$ -equivalence. Thus, if $r < m-r-1$, we have both $\pi_r(M') \cong \pi_r(W)$ and $\pi_r(W) \cong \pi_r(M)/\langle\langle [f] \rangle\rangle$. The result follows. $\square$

It turns out that given a B -structure we have good control over which embeddings $S^r \hookrightarrow M$ we can do surgery on. Specifically, if $r < \frac{m}{2}$, then the Whitney embedding theorem ([20] Theorem 4.11) gives that any map $S^r \hookrightarrow M$ is homotopic to an embedding f . If $\bar{\nu}_*([f]) = 0$ then of course $\nu_*([f]) = 0$ and thus $[f]$ has stably trivial normal bundle. It turns out that in this range of dimensions this actually implies that $[f]$ has immediately trivial normal bundle. This uses the following lemma:

Lemma 3.2. *([8] p.169) Let B be a cell complex, and G a framing of a rank n vector bundle $E \rightarrow B$. Given any k -frame field G_1 in E with $\dim B < n-k$, there is an $n-k$ frame field in E such that $G_1 + G_2$ is homotopic to G .*

Proof. We may assume all framings are orthonormal (See [8]). Fix a framing F of E . For each $x \in M$, let $h_F(G)(x) \in O(n)$ be the base change matrix between the basis determined by F at x and that determined by G . We have the (not necessarily commutative) diagram

$$\begin{array}{ccc} & & O(n) \\ & \nearrow h_F(G) & \downarrow p \\ B & \xrightarrow{h_F(G_1)} & O(n)/O(n-k) \end{array}$$

Where p is a fibration. Since $O(n)/O(n-k)$ is $(n-k-1)$ -connected ([8]), and $\dim B < n-k$, $p \circ h_F(G)$ and $h_F(G_1)$ are homotopic as they are both homotopic to a constant map. Since p is a fibration there is thus a map $h' : B \rightarrow O(n)$ with $h' \simeq h_F(G)$ such that $p \circ h' = h_F(G_1)$. Clearly, h' determines a framing G' of E by just considering the pointwise image $h'(x)(F_x)$. Let G_2 be the $(n-k)$ -frame field given by the last $n-k$ sections of G' . Then, $G' = G_1 + G_2$ and $G' \simeq G$, as required. $\square$

Now, suppose $E \rightarrow S^r$ is a stably trivial bundle with fiber dimension k greater than r . Say $E \oplus \epsilon^i$ is trivial. Let G_i be an orthonormal framing of ϵ^i . By the above lemma, since $k+i-i > r$, there is a k -frame field G in $E \oplus \epsilon^i$ such that $G + G_i$ is a framing of $E \oplus \epsilon^i$. Then G is a framing of E and thus E is trivial.

In particular, when $r < \frac{m}{2}$, the dimension of the normal bundle of an embedding $S^r \hookrightarrow M^m$ is $m-r > r$, so a stably trivial normal bundle is just trivial in this case. Thus we can do surgery to kill elements in $\ker(\bar{\nu}_* : \pi_r(M) \rightarrow \pi_r(B))$ in this range.

We want to show that we can perform such surgeries while also maintaining control of a new B -structure on the resulting manifold. The following lemma gives the required information

Lemma 3.3. *Let $r > 0$, and $f : S^r \hookrightarrow M$ be an embedding with $r < \frac{m}{2}$ and such that $\bar{\nu}_*([f]) = 0$. Then, there exists a thickening $\bar{f} : S^r \times D^{m-r} \hookrightarrow M$ such that $\bar{\nu} : M \rightarrow B$ extends to a normal B -structure of W , the trace of surgery via $\bar{f}$.*

Proof. Since $\bar{\nu}_*([f]) = 0$, there exists some trivialization $n : D^{r+1} \rightarrow B$ fitting into the commutative diagram

$$\begin{array}{ccc} D^{r+1} & \xrightarrow{n} & B \\ \uparrow \iota & \nearrow \bar{\nu} \circ f & \downarrow \theta \\ S^r & \xrightarrow{\nu \circ f} & BO \end{array}$$

But of course we can also consider the fibration

$$O/O(m-r) \rightarrow BO(m-r) \rightarrow BO$$

We have a lift ν_f of the map $\nu \circ f : S^r \rightarrow BO$ to $BO(m-r)$ corresponding to a tubular neighborhood of $[f]$. We then want to lift the trivialization $\theta \circ n : D^{r+1} \rightarrow BO$ to $BO(m-r)$ relative to the lift ν_f of

$S^r = \partial D^{r+1}$. The obstruction to doing this lift lives in $\pi_{r+1}(BO, BO(m-r)) \cong \pi_r(O/O(m-r))$, but $O/O(m-r)$ is $(m-r-1)$ -connected, so since $m-r > r$ this group vanishes, so such a lift must exist. This is summarized in the commutative diagram

The map t gives a trivialization of ν_f , so specifically it gives an embedding $\bar{f} : S^r \times D^{m-r} \hookrightarrow M$, and then n gives a B -structure on the core of the handle that will be glued on. This B -structure and the original one on M glue together because $n \circ \iota = \bar{\nu} \circ f$.

□

Note that this goes through in the $r = 0$ case if B is also assumed connected.

Lemma 3.4. *Suppose M is a B -manifold of dimension $m \geq 4$, where B is connected with a finite $[m/2]$ skeleton. By a finite sequence of surgeries, we can replace $(M, \bar{\nu})$ by $(M', \bar{\nu}')$ such that $\bar{\nu}' : M' \rightarrow B$ is an $[m/2]$ -equivalence.*

Proof. First, we can surger M by finitely many $S^0 \times D^m$'s to connect it (This is why we must assume B is connected: It allows us to use the remarked strengthening of Lemma 3.3 to extent the B -structure to the trace of this surgery).

Now, we want to force the induced map $\bar{\nu}_*$ on π_{i+1} to be surjective. We are essentially going to connect-sum sphere bundles onto M which are equipped with B -structures that land on the correct elements of $\pi_{i+1}(B)$. Specifically, given a class $A \in \pi_{i+1}(B)$, let $\alpha \in \pi_{i+1}(BO)$ be the image of A under the induced map from the fibration. Since we are assuming $i+1 \leq [m/2]$, the induced map of the fibration $BO(m-i-1) \rightarrow BO$ on π_{i+1} is surjective. Pick some $\bar{\alpha} : S^{i+1} \rightarrow BO(m-i-1)$ in the preimage of α . Let $V \rightarrow S^{i+1}$ be the corresponding vector bundle and $X_{\bar{\alpha}} := S(V)$, the sphere bundle of V , with projection $\pi : E_{\bar{\alpha}} \rightarrow S^{i+1}$.

The Gysin sequence gives

$$0 \rightarrow H^{i+1}(S^{i+1}) \rightarrow H^{i+1}(X_{\alpha}) \rightarrow H^{2i+3-m}(S^{m-i-1}) \cong 0$$

Thus the base sphere of the bundle X_{α} represents a free homotopy class $S \in \pi_{i+1}(X_{\alpha})$. It is a standard property of sphere bundles that, stably,

$$TX_{\bar{\alpha}} \simeq \pi^*TS^{i+1} \oplus \pi^*V$$

Of course, as the tangent bundle of S^{i+1} is stably trivial, this means $TX_{\bar{\alpha}} \simeq \pi^*V$. Then, since

$$TS^{i+1} \oplus \nu_{S^{i+1} \subset X_{\bar{\alpha}}} \cong s^*TX_{\bar{\alpha}}$$

(where $s : S^{i+1} \rightarrow X_{\bar{\alpha}}$ is the zero section), we get that $\nu_{S^{i+1} \subset X_{\bar{\alpha}}} \simeq s^*\pi^*V \cong V$ stably, and thus $(\nu_{X_{\alpha}})_*(S) = \alpha \in \pi_{i+1}(BO)$.

We can then pick a lift $\bar{\nu} : X_\alpha \rightarrow B$ such that $\bar{\nu}(S) = A$. This gives a well-defined B -structure on $M \coprod X_\alpha$, which by the previous lemma induces one on $M \# X_\alpha$, the result of a surgery via $S^0 \times D^m$. This B -structure clearly contains A in its image. Since $\pi_1(B)$ is finitely generated and $\pi_i(B)$ is finitely generated over $\mathbb{Z}[\pi_1(B)]$ for $i > 1$, by a finite sequence of these surgeries we can make $\bar{\nu}_*$ surjective on π_i for $i \leq m/2$ without changing anything in lower dimensions.

We now force injectivity. Suppose $\langle x_1, \dots, x_k | r_1, \dots, r_s \rangle$ is a presentation of $\pi_1(B)$. By our previous discussion we can connect-sum M with various sphere bundles to get an $M' := M \# X$ such that $\pi_1(M')$ has presentation

$$\langle a_1, \dots, a_j, z_1, \dots, z_k | R'_1, \dots, R'_p \rangle$$

such that $\langle a_1, \dots, a_j | R'_1, \dots, R'_p \rangle \cong \pi_1(M)$, $\langle z_1, \dots, z_k \rangle \cong \pi_1(X)$, and $\bar{\nu}'_*(z_i) = x_i$. To get injectivity, since we have better control of where the generators $\{z_i\}$ are sent by $\bar{\nu}'_*$, we'd first like to surger M' to have $\pi_1(M'')$ generated by just $\{z_i\}$. We must do this by relating each a_i with $\{z_i\}$, and in our context there is an easy method. Specifically, writing $\bar{\nu}'_*(a_i) = w_i(x_1, \dots, x_k)$, a word in $\{x_i\}$, clearly $\bar{\nu}'_*(a_i^{-1}w_i(z_1, \dots, z_k)) = 0$, and thus we can surger M' to add this as a relation, since the homotopy class represented by this word has stably trivial normal bundle and thus immediately trivial normal bundle as the dimension is less than $m/2$. This gives a manifold M'' with presentation $\langle z_1, \dots, z_k | \{R''_i\} \rangle$. It is clear that $\{R''_i\}$ considered as words in $\{x_i\}$ is generated by $\{r_i\}$ (by the correspondence given by $\bar{\nu}''$) so we just need the opposite inclusion to also hold. That is, we want to add $\{r_i(z_1, \dots, z_k)\}$ as relators. Again, since these are in the kernel of $\bar{\nu}''_*$ we can do this, and the resulting $\bar{\nu}'''$ is an isomorphism on π_1 .

We can thus assume that M is connected and $\bar{\nu} : M \rightarrow B$ is 2-connected. We proceed inductively. That is, suppose $\bar{\nu}$ is r -connected for $2 \leq r < [m/2]$. $\bar{\nu}$ is homotopy equivalent to an embedding $M \hookrightarrow B$, so the LES on relative homotopy gives

$$\pi_{r+1}(B, M) \xrightarrow{d} \pi_r(M) \xrightarrow{\bar{\nu}_*} \pi_r(B) \rightarrow 0$$

Since B has a finite $(r+1)$ -skeleton, $H_{r+1}(B, M; \Lambda) \cong \pi_{r+1}(B, M)$ is finitely generated, by the relative Hurewicz theorem. We can thus surger M to kill the image of d (the kernel of $\bar{\nu}_*$). We can then connect-sum by sphere bundles to make the induced map on π_{r+1} surjective. □

Note that by the proof of this lemma, the modified manifold is B -bordant to the original one. We now have enough to prove the stabilization theorem.

Proof. (of Theorem) We're given a B -bordism (W, M_1, M_2) . By the previous lemma, we may assume that $\bar{\nu}_W$ is a q -equivalence. Now, by definition of a B -bordism, the diagram

$$\begin{array}{ccc} W & \xrightarrow{\bar{\nu}_W} & B \\ \uparrow \iota & \nearrow \pm \bar{\nu}_{M_i} & \\ M_i & & \end{array}$$

commutes, so by functoriality of H_k the diagram

$$\begin{array}{ccc}
H_k(W) & \xrightarrow{(\bar{\nu}_W)_*} & H_k(B) \\
\uparrow \iota_* & \nearrow (\pm \bar{\nu}_{M_i})_* & \\
H_k(M_i) & &
\end{array}$$

does too. By assumption, for $i = 1, \dots, q-1$ $(\bar{\nu}_W)_*$ and $(\bar{\nu}_{M_i})_*$ are isomorphisms, so ι_* in those degrees must be, too. The long exact sequence for relative homology for the pair (W, M_i) then gives that $H_k(W, M_i) \cong 0$ for $i = 1, \dots, q-1$. By the Normal Form Lemma, we thus have a decomposition of W into handles of the form

$$W \cong M_0 \times [0, 1] + \sum_{i=1}^{p_q} (\phi_i^q) + \sum_{i=1}^{p_{q+1}} (\phi_i^{q+1})$$

Note then that the attaching spheres of the q -handles are nullhomotopic in W , and hence also in M_i (by the above commuting triangles). This has the effect of connect-summing $S^q \times S^q$'s to M_0 (note that since they are nullhomotopic we can assume the attaching spheres of the q -handles live in some Euclidean neighborhood of $M_0 \times \{1\}$, whereupon noticing that disjoint copies of S^{q-1} in $\mathbb{R}^{2q}$ cannot be nontrivially linked gives this result). We can then consider the dual handle decomposition and get a similar operation on M_1 . The result follows. $\square$

4 Cancellation of $S^n \times S^n$'s

We now study when a diffeomorphism $f : M \# r(S^n \times S^n) \rightarrow N \# r(S^n \times S^n)$ actually implies a diffeomorphism $M \xrightarrow{\cong} N$. We note first that by the theorem of Haefliger, when M is simply-connected every element of $\pi_n(M \# r(S^n \times S^n))$ may be represented by a smooth embedding of S^n . Thus there is a well-defined $(-1)^n$ -symmetric intersection form on $\pi_n(M \# r(S^n \times S^n))$. The condition for destabilization will involve this, as well as a quadratic refinement μ which essentially attempts to record self-intersection data. However, this is only actually well-defined if we restrict to the module $K\pi_n(M) = \ker(\nu_* : \pi_n(M) \rightarrow \pi_n(BO))$ (See the discussion in [10] Section 3).

Note that every element of $\pi_n(S^n \times S^n)$ is represented by an embedding with trivial normal bundle (because $S^n \times \{0\}$ has the tubular neighborhood $S^n \times (S^n - \{\infty\})$), so the intersection form and the extra invariant α is always defined on all of $\pi_n(\#r(S^n \times S^n))$. The diffeomorphism f induces the map

$$\mathbb{Z}^{2r} \cong \pi_n(\#r(S^n \times S^n)) \rightarrow K\pi_n(M) \oplus \pi_n(\#r(S^n \times S^n)) \xrightarrow{f_*} K\pi_n(N) \oplus \pi_n(\#r(S^n \times S^n)) \rightarrow \pi_n(\#r(S^n \times S^n)) \cong \mathbb{Z}^{2r}$$

denoted $\vartheta(f)$. The condition we will need turns out to be that $\vartheta(f)$ preserves the restricted intersection form. Of course, the first two maps in the above sequence are obviously isometries, so the condition is really dependent on the final projection to the second factor.

Theorem 4.1. *Let $n > 1$. Suppose $\pi_1(M) \cong \{1\}$ and that there exists a diffeomorphism*

$$f : M \#_r (S^n \times S^n) \rightarrow N \#_r (S^n \times S^n)$$

If $\vartheta(f)$ is an isometry with respect to the intersection form (λ, μ) , then, M and N are diffeomorphic via a diffeomorphism extending $f|_{\partial M} : \partial M \rightarrow \partial N$.

The strategy of proving this will be to construct a cobordism between M and N by forming two cylinders $M \times [0, 1]$, $N \times [0, 1]$, attaching $\natural_r(S^n \times D^{n+1})$ to $M \times \{1\}$ and $\natural_r(D^{n+1} \times S^n)$ to $N \times \{0\}$ and then gluing these two pieces together via the diffeomorphism f . It will turn out that, as long as we can control $\vartheta(f)$ sufficiently, we can force this to be an h -cobordism. To get this control, we must first discuss when we can induce isometries of the intersection form on $\pi_n(\#_r(S^n \times S^n))$ by diffeomorphisms.

4.1 Automorphs of the intersection form

The purpose of this subsection is to prove the following lemma, which will be key in destabilizing a stable diffeomorphism.

Lemma 4.1. *Write $W_r := \#_r(S^n \times S^n)$. Then there exists $s \in \mathbb{N} \cup \{0\}$ such that for any automorphism*

$$A : H_n(W_{r+s}) \rightarrow H_n(W_{r+s})$$

which preserves λ and μ , there exists a self-diffeomorphism g of W_{r+s} such that $g_ = A$.*

The cases $n = 2$ and $n \geq 3$ behave somewhat differently. The $n = 2$ case is done more or less by hand in [22]. We give a sketch of the ideas here.

Given any simply connected 4-manifold N (Which we will later take to be W_r), form the connect-sum $M := N \# (S^2 \times S^2)$ via surgery on an embedded $S^1 \hookrightarrow N$ with trivial normal bundle. We then define a self-diffeomorphism of M by isotoping the connecting S^1 around N , tracing out an element $\omega \in H_2(N)$. Since $\pi_2(SO(3)) \cong \mathbb{Z}_2$, when $\omega^2 = 2m$, the trivial normal bundle of S^1 is mapped back to itself and twisted in such a way as to still have the result of a surgery along it be connect-summing with $S^2 \times S^2$. We thus get an induced self-diffeomorphism g of M . If $x, y \in H_2(S^2 \times S^2) < H_2(M)$ are the hyperbolic generators and $u \in H_2(N) < H_2(M)$ is any element, it turns out that this map satisfies

$$g_*(u) = u - \langle u, \omega \rangle y, \quad g_*(x) = x + \omega - sy, \quad g_*(y) = y$$

Such a map is called a transvection. Now, assuming the intersection form on $H_n(N)$ is indefinite (that is neither positive definite nor negative definite), and also even (that is $\lambda(x, x) \in 2\mathbb{Z}$ for all x), it is an algebraic fact ([25]) that these transvections, along with the induced map on $H_n(M)$ of the diffeomorphism of swapping the coordinates of $S^2 \times S^2$, generate any automorph of $H_n(M)$. Of course, the intersection form on W_r has $\lambda(e_1 + f_1, e_1 + f_1) = 2$, $\lambda(e_1 - f_1, e_1 - f_1) = -2$ and is also clearly even. Thus Lemma 4.1 is proved for $n = 2$.

For $n \geq 3$, we essentially follow [23]. In the $(n - 1)$ -connected case, preserving μ is equivalent to preserving the map $\alpha : \pi_n(W_r) \rightarrow \pi_{n-1}(SO(n))$ classifying the normal bundle of elements of π_n . Note that this map is well-defined when $n \geq 4$ because homotopic embeddings are diffeotopic and of course diffeomorphisms preserve the normal bundle of embedded submanifolds. The map α is well-defined when $n = 3$ because $\pi_2(SO(3)) \cong \{1\}$.

From now on write $N := W_r - \text{int}(D)$. It is a fact that we may construct N by attaching $2r$ disjoint copies of $D^n \times D^n$ via embeddings $f_i : \partial D^n \times D^n \rightarrow \partial D^{2n}$. (To illustrate the construction it is easy

when $r = n = 1$. We split $S^1 = \partial D^2$ into quarters and glue the components of each $S^0 \times D^1$ onto diametrically opposite quarters). Such a decomposition is called a *presentation* for N . Note that given such a presentation, $f_i(\partial D^n \times \{0\})$ bounds a disc in D^{2n} (simply because D^{2n} is contractible), and it also bounds $D^{n-1} \times \{0\}$. Gluing these discs together we get an element of $\pi_n(N)$, and all such spheres together form a basis.

It will be key that there in fact exists a presentation of N corresponding to *any* given basis of $\pi_n(N)$. We will show this using the h -cobordism theorem:

We want to make precise the following process: Take embeddings $\bar{f}_i : S^n \hookrightarrow N$ representing a given basis of $H_n(N)$. Let $D' \subset N$ be some embedded disc. We may assume that the images of f_i intersect D' in some small copies of D^n , and that these images only intersect each other in the interior of D' . Thus cutting out D' from N gives disjointly embedded D^n 's with boundaries on $\partial D'$. Thicken these embedded D^n 's to disjoint tubular neighborhoods. This gives a presentation of some manifold with these generators. But this process turns out to generate an h -cobordism which shows that this manifold is just N .

Doing this precisely, write $K := \overline{N - D'}$ for some disc D' in the interior of N . By excision, $H_n(N, D')$ and $H_n(K, \partial D')$ are isomorphic, and by the Hurewicz theorem this latter group is isomorphic to $\pi_n(K, \partial D')$. Thus $\pi_n(K, \partial D') \cong H_n(N)$, so we can represent any given basis of $H_n(N)$ by maps $\bar{f}_i : (D^n, \partial D^n) \rightarrow (K, \partial D')$. We may assume these are embeddings, and that the images of these maps only intersect in D' since W_r is 2-connected (this can be done using methods from [19]). We can then thicken these to tubular neighborhoods. Of course, the tubular neighborhood of a contractible submanifold is trivial, so we get maps $f_i : (D^n, \partial D^n) \times D^n \rightarrow (K, \partial D')$. We take $N' = D' \bigcup_i \text{im}(f_i)$. By Whitehead's theorem, the inclusion $N' \hookrightarrow N$ is a homotopy equivalence. This construction of course gives a presentation of N' by considering the restrictions $\{f_i|_{\partial D^n \times D^n}\}$.

Let $L := \overline{N - N'}$. This will turn out to be an h -cobordism between ∂N and $\partial N'$. Indeed, by excision $H_*(L, \partial N') \cong H_*(N, N')$ and this is 0 as $N' \hookrightarrow N$ is a homotopy equivalence. Lefschetz Duality gives that $H_*(L, \partial N) \cong 0$. Thus ∂N and $\partial N'$ are diffeomorphic, and L is diffeomorphic relative to ∂N to $\partial N \times [0, 1]$. This means that N is obtained from N' by gluing on a collar $\partial N' \times [0, 1]$, and thus N and N' are diffeomorphic. This gives a presentation of N with the correct homological data.

Then, [23] shows that any two presentations with the same invariant data (that is, the same intersection/self-intersection data) may be sent to each other by a diffeomorphism of N fixing its boundary. Given an automorph A of $H_n(N)$, bases $\{e_i\}$ and $\{A(e_i)\}$ represent presentations that do share the same intersection data, so these are sent to each other by a diffeomorphism g which clearly then induces A on H_n .

4.2 Proof of the destabilization theorem

Proof. (Of Theorem 4.1)

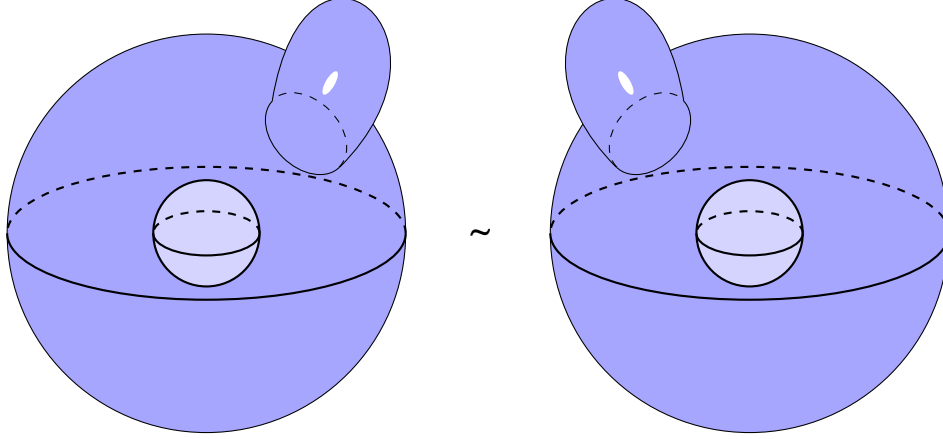
First note that we can assume that $\vartheta(f) = \text{id}$. This is because by Lemma 4.1 there exists a diffeomorphism $g : M \#(r+s)(S^n \times S^n) \rightarrow M \#(r+s)(S^n \times S^n)$ such that $\vartheta(g) = \vartheta(f) \oplus \text{id}$, and g induces the identity on $H_n(M)$. Then, $f' := (f \# \text{id}) \circ g^{-1}$ has $\vartheta(f') = \text{id}$.

We now consider the following bordism:

$$[M \times [0, 1] \natural r(S^n \times D^{n+1})] \cup_f [N \times [0, 1] \natural r(D^{n+1} \times S^n)]$$

Where $\natural$ is the boundary connected sum, $\partial(S^n \times D^{n+1})$ is being glued onto $M \times \{1\}$, $\partial(D^{n+1} \times S^n)$ is

being glued onto $N \times \{0\}$, and f is considered a map between the two resulting copies of $M \#_r(S^n \times S^n)$ and $N \#_r(S^n \times S^n)$. We will write this as $X \cup_f Y$. To illustrate this construction in a lower dimension, if $M \cong N \cong S^2$, we have the identity diffeomorphism $f : M \#(S^1 \times S^1) \rightarrow N \#(S^1 \times S^1)$. The cobordism in this case is:



where the outer boundaries are copies of $S^2 \#(S^1 \times S^1)$ glued together via the diffeomorphism. This of course just leaves the two interior copies of S^2 as cobordant manifolds.

Returning now to the general case, this turns out to be an h -cobordism. Indeed, considering the triple $(X \cup_f Y, Y, N)$ gives the following long exact sequence

$$\dots \rightarrow H_{n+1}(X \cup_f Y, N) \rightarrow H_{n+1}(X \cup_f Y, Y) \rightarrow H_n(Y, N) \rightarrow H_n(X \cup_f Y, N) \rightarrow \dots$$

By excision, $H_*(X \cup_f Y, Y) \cong H_*(X, M \#_r(S^n \times S^n))$. But collapsing the cylinder of X we see that it deformation retracts onto $M \times \{1\}$ attached to $\natural_r(S^n \times D^{n+1})$ in the clear way induced by our construction.

This space, $X - M \times [0, 1)$ (Call it L), looks like the connect-sum of $\#_r(S^n \times S^n)$ with M , but with the connecting disc *filled in* after it is removed to perform the connect sum, and then also the cavity in the $\#_r(S^n \times S^n)$ component is filled in. Denote by D the connecting disc which is added back to form L .

By excision again we get $H_*(X \cup_f Y, Y) \cong H_*(L - (M - D), \#_r(S^n \times S^n) - \text{int}(D))$. What is really happening is we have split the boundary of $L - (M - D) \cong \natural_r(S^n \times D^{n+1})$ into two components D and $\#_r(S^n \times S^n) - \text{int}(D)$, and are calculating the relative homology of $\natural_r(S^n \times D^{n+1})$ paired with the latter. But by Lefschetz duality for triples, we can instead pair this space with the former boundary to get that $H_*(X \cup_f Y, Y) \cong H^{n+1-*}(\natural_r(S^n \times D^{n+1}), D) \cong \tilde{H}^{n+1-*}(\natural_r(S^n \times D^{n+1})) \cong \tilde{H}^{n+1-*}(\bigvee_r S^n)$.

Also, it is easier to see that by construction $H_*(Y, N) \cong H_*(\natural_r(S^n \times D^{n+1}), \text{disc}) \cong \tilde{H}_*(\bigvee_r S^n)$. We thus get the exact sequence

$$\mathbb{Z}^r \xrightarrow{\partial} \mathbb{Z}^r \rightarrow H_n(X \cup_f Y, N) \rightarrow 0$$

where ∂ is the restriction of $\vartheta(f)$ to $(\{0\} \times \mathbb{Z})^r$. Now, because we forced $\vartheta(f) = \text{id}$, this is immediately an isomorphism, and thus $H_*(X \cup_f Y, N) \cong 0$.

□

It turns out we can immediately ensure that $\vartheta(f)$ is an isometry by controlling the intersection form on $K\pi_n(M) = \ker(\nu_* : \pi_n(M) \rightarrow \pi_n(BO))$ and $K\pi_n(N)$.

Lemma 4.2. *Suppose $\pi_1(M) \cong \{1\}$ and that the intersection form vanishes on $K\pi_n(M)$ and $K\pi_n(N)$. Then for any diffeomorphism $f : M \#_r(S^n \times S^n) \rightarrow N \#_r(S^n \times S^n)$, $\vartheta(f)$ is an isometry.*

Proof. The intersection form is defined on $K\pi_n(M \#_r(S^n \times S^n)) = K\pi_n(M) \oplus \pi_n(\#_r(S^n \times S^n))$. Let $x, y \in \pi_n(\#_r(S^n \times S^n))$ be arbitrary. Suppose $f_*(x) = a_x + b_x$, $f_*(y) = a_y + b_y$, with $a_{x,y} \in K\pi_n(N)$, $b_{x,y} \in \pi_n(\#_r(S^n \times S^n))$. Then, $\lambda(\vartheta(f)(x), \vartheta(f)(y)) = \lambda(\text{proj}_2 \circ f_*(x), \text{proj}_2 \circ f_*(y)) = \lambda(b_x, b_y)$, but since λ is 0 on $K\pi_n(M)$, this is equal to $\lambda(a_x + b_x, a_y + b_y) = \lambda(f_*(x), f_*(y)) = \lambda(x, y)$. Hence, $\vartheta(f)$ preserves λ . □

5 An example: $\mathbb{H}\mathbb{P}^2$ and an exotic sphere

In this section we apply some of the tools we've developed to construct an exotic 8-sphere Σ which we show has the property that $\mathbb{H}\mathbb{P}^2 \# \Sigma$ is again diffeomorphic to $\mathbb{H}\mathbb{P}^2$. We briefly recall that the Whitehead tower of BO begins as

$$\dots \rightarrow BString \rightarrow BSpin \rightarrow BSO \rightarrow BO$$

and that the obstruction to lifting the stable normal Gauss map $\nu : M \rightarrow BO$ to BSO , $BSpin$, and $BString$ is respectively the first Stiefel-Whitney class $w_1(TM)$, the second Stiefel-Whitney class, and the first Pontryagin class $p_1(TM)$.

Begin by noting that $H^i(S^8; \mathbb{Z}_2) \cong 0$ for $i = 1, 2$ and $H^4(S^8; \mathbb{Z}) \cong 0$. Thus $w_i(TS^8) = 0$ for $i = 1, 2$ and $p_1(TS^8) = 0$ so the classifying map $\nu : S^8 \rightarrow BO$ lifts to a $\bar{\nu} : S^8 \rightarrow BString$. By [5] p.538, the bordism group $\Omega_8^{\text{string}} \cong \mathbb{Z} \oplus \mathbb{Z}_2$, where the first coordinate is determined by the signature of a given manifold. Let M then be some manifold in the class $(0, 1) \in \Omega_8^{\text{string}}$.

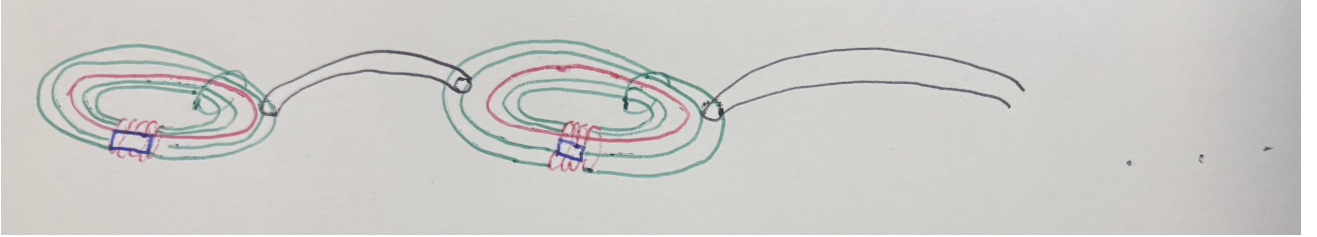
By Whitney's embedding theorem (the version we've been using), every element of $\pi_i(M)$ for $1 \leq i \leq 3$ can be represented by an embedding. Since M has a string structure, all these embeddings may be assumed to have trivial normal bundle, as $\pi_{1,2,3}(BString) \cong 0$, so $\nu : \pi_{1,2,3}(M) \rightarrow \pi_{1,2,3}(BO)$ factors through 0. This means the embeddings have stably trivial normal bundle, and we are in the range of dimensions where Lemma 3.2 shows these must be immediately trivial. Thus we can surger to get an M' with $\pi_i(M') = 0$ for $1 \leq i \leq 3$, and thus also $5 \leq i \leq 7$ by Poincaré Duality. Note that we may perform these surgeries without changing the $BString$ -bordism class.

We now consider the middle homology. Since $[M'] = (0, 1) \in \Omega_{String}^8$, the signature of the intersection form on M' is 0. Now, note that the map $H^n(M'; \mathbb{Z}_2) \rightarrow \mathbb{Z}_2$ given by $x \mapsto \langle x \smile x, [M'] \rangle$ is linear, and is thus by Poincaré Duality represented by cupping with some $v \in H^n(M'; \mathbb{Z}_2)$, called the *Wu Class* of M' . However, it is a fact ([16] Theorem 11.14) that since M' has a spin structure, $w = \frac{1}{2}p_1$. Of course since M' has a string structure, this means $w = 0$. Hence, the intersection form on $H_k(M')$ is even, and by [17] Chapter II Theorem 5.3, unimodular indefinite symmetric bilinear forms are classified over the integers by their signature, rank, and parity. Thus, the intersection form is equivalent to

$$\bigoplus_{g=1}^k \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

Now, let $\{e_1, f_1, \dots, e_k, f_k\} \subseteq H_4(M')$ be a basis such that $e_i \cdot f_i = 1$ and products are 0 otherwise. Since M' is 3-connected, these can be represented by maps $S^4 \rightarrow M'$ which by the theorem of Haefliger can be

assumed to be embeddings. By using the Whitney trick (valid as $2 \dim S^4 = \dim M'$), we can guarantee they intersect exactly as many times as the intersection form requires, that is the spheres representing e_i and f_i intersect transversally exactly once, and all other spheres are disjoint. Since $\pi_4(BString) \cong 0$, these embeddings have stably trivial normal bundles. Let $N \rightarrow S^4$ be one of these bundles. Now, $\pi_3(SO(4)) \cong \mathbb{Z} \oplus \mathbb{Z}$, so rank 4 bundles on S^4 are classified by the first Pontryagin class p_1 , and the Euler class e [16]. By the additivity of p_1 we have $0 = 2p_1(\epsilon^{4+i}) = 2p_1(N \oplus \epsilon^i) = 2p_1(N) + 2p_1(\epsilon^i)$, thus $p_1(N) = 0$. For the Euler class, this is just the self-intersection number which by assumption was 0. Thus each of our embeddings has actually trivial normal bundle, so we have k pairs of embedded $S^4 \times D^4$'s with middle circles that intersect transversely once and which are disjoint from other pairs. This gives k disjoint embeddings of $S^4 \times D^4 \cup D^4 \times S^4 \cong S^4 \times S^4 - D^8$. Using a similar idea as when proving the Modification Lemma, we can connect these submanifolds by tubes which are disjoint from them besides some small connecting S^4 's. Doing this, we get that $\#_k[(S^4 \times S^4) - D^8]$ embeds into M' . Let its image in M' be L .



The long exact sequence on relative homology gives

$$0 \rightarrow H_5(M'/L) \rightarrow H_4(L) \xrightarrow{\iota_*} H_4(M') \rightarrow H_4(M'/L) \rightarrow 0$$

By construction, ι_* is an isomorphism in this degree. Thus $H_4(M'/L) \cong H_5(M'/L) \cong 0$. We also get

$$0 \rightarrow H_8(M') \rightarrow \tilde{H}_8(M'/L) \rightarrow H_7(L) \rightarrow 0$$

Of course $H_7(L) \cong \mathbb{Z}^{k-1}$ so this splits and we get that $\tilde{H}_*(M'/L) \cong \tilde{H}_*(\bigvee_{i=1}^k S^8)$.

Hence, $M' \cong \Sigma_k \# \Sigma_{k-1} \# \dots \# \Sigma_1 \# k(S^4 \times S^4)$ for some homotopy spheres $\{\Sigma_i\}$. Now, we see that S^8 and $S^4 \times S^4$ are *BString*-bordant by considering the trace of the surgery on S^8 via an embedded $S^3 \times D^5 \hookrightarrow S^8$, and using Lemma 3.3. In general, the connect-sum of two manifolds is cobordant to their disjoint union, by considering a connecting surgery of $S^0 \times D^m$, and using the lemma that lets us extend the *B*-structure to the trace of the surgery. Thus $[\Sigma_k] + [\Sigma_{k-1}] + \dots + [\Sigma_1] = [M'] = (0, 1) \in \Omega_8^{\text{string}}$ so at least one $\Sigma_j =: \Sigma$ is not diffeomorphic to S^8 . Note regardless that as Σ is a homotopy 8-sphere it is actually homeomorphic to S^8 , by the Poincaré conjecture in this dimension (which we showed to be true).

Now, we show that $\mathbb{H}\mathbb{P}^2 \# \Sigma \cong \mathbb{H}\mathbb{P}^2$. We do this by showing that these manifolds have bordant normal $(4-1)$ -types, thus giving us a stable diffeomorphism. We then show that we can destabilize.

Since $H^i(\mathbb{H}\mathbb{P}^2; \mathbb{Z}_2) \cong 0$ for $i = 1, 2$ (just by the cell decomposition), it is spin. Note that by [1], $\Omega_8^{\text{spin}} \cong \mathbb{Z} \oplus \mathbb{Z}$.

Within Ω_8^{spin} , $[\mathbb{H}\mathbb{P}^2 \# \Sigma] = [\mathbb{H}\mathbb{P}^2] + [\Sigma]$. Now note that $\Sigma \# \Sigma$ is *BString*-bordant to S^8 , say via a bordism W . Since *BString* has a finite 5-skeleton, this bordism can be modified so that $\bar{\nu}_W$ is a

5-equivalence. However then W is clearly an h -cobordism, thus $\Sigma \# \Sigma$ is diffeomorphic to S^8 . Hence, $[\Sigma] = (0, 0) \in \Omega_8^{\text{spin}}$ as this is the only torsion element.

Thus, $\mathbb{HP}^2 \# \Sigma$ and $\mathbb{HP}^2$ are $BSpin$ bordant. To apply Kreck's stable diffeomorphism theorem, we want to show that $\bar{\nu}_{\mathbb{HP}^2} : \mathbb{HP}^2 \rightarrow BSpin$ is a 3-smoothing. Suppose $\pi_4(\mathbb{HP}^2)$ is generated by some embedding $S^4 \hookrightarrow \mathbb{HP}^2$ with image S . Since $\mathbb{HP}^2$ admits a Riemannian metric (say by pulling back the inclusion $\mathbb{HP}^2 \hookrightarrow \mathbb{R}^N$ into some high-dimensional euclidean space), the exact sequence

$$0 \rightarrow TS \rightarrow T\mathbb{HP}^2|_S \rightarrow N_{S/\mathbb{HP}^2} \rightarrow 0$$

of bundles splits. That is, $T\mathbb{HP}^2|_S \cong TS \oplus N_{S/\mathbb{HP}^2}$. Then, since TS^4 is stably trivial and by additivity of p_1 , we get that $p_1(N_{S/\mathbb{HP}^2}) = p_1(T\mathbb{HP}^2|_S)$, but this latter value is just the pullback $\iota^* p_1(T\mathbb{HP}^2)$ along the inclusion $\iota : S \rightarrow \mathbb{HP}^2$. Of course, by definition of S , ι^* is an isomorphism on π_4 . Now, by ([16] p. 248), $p_1(T\mathbb{HP}^2) = 2$. By problem 15-A in [16], the first Pontryagin class of a bundle over S^4 is even, so we must have that $\nu_*([S]) = 1 \in \pi_4(BO)$ and hence $\bar{\nu}_*$ is an isomorphism on π_4 . Hence it is a 3-smoothing.

We thus have a diffeomorphism $\mathbb{HP}^2 \# \Sigma \# r(S^4 \times S^4) \cong \mathbb{HP}^2 \# r(S^4 \times S^4)$ for some r . Since by the previous discussion $\ker(\nu_* : \pi_4(\mathbb{HP}^2) \rightarrow \pi_4(BO)) \cong \{0\}$, by Lemma 4.2 $\mathbb{HP}^2 \# \Sigma$ and $\mathbb{HP}^2$ are diffeomorphic. This is a special case of the following result which was proved in [12]

Theorem 5.1. *Suppose M is a closed simply-connected smooth manifold with $\dim M = 2m > 5$, and such that $H_*(M) \cong \mathbb{Z} \oplus \mathbb{Z} \oplus \mathbb{Z}$. For any homotopy $2m$ -sphere Σ , $M \# \Sigma$ is diffeomorphic to M .*

The simplest examples of such manifolds are $M = \mathbb{HP}^2, \mathbb{OP}^2$. We can in fact do the $\mathbb{OP}^2$ case with almost the same argument as $\mathbb{HP}^2$: By [18] p.504 there is a single exotic 16-sphere Σ and this has $\Sigma \# \Sigma \cong S^{16}$. Since $H_4(\mathbb{OP}^2) \cong 0$, $p_1(T\mathbb{OP}^2) = 0$ so $\mathbb{OP}^2$ has a string structure. This gives the normal 7-type of $\mathbb{OP}^2$ as $BString$ is 7-connected, and the lift is a 7-smoothing as $\pi_8(BString) \cong \mathbb{Z}$, and by [7], $p_2(T\mathbb{OP}^2) \cong 6u$ (By [16] p.244, stable oriented 8-dimensional vector bundles over S^8 are classified by $\frac{p_2}{6}$). Since $[\Sigma] \in \Omega_{\text{String}}^{16} \cong \mathbb{Z} \oplus \mathbb{Z}$ is torsion, it is 0. This gives a stable diffeomorphism between $\mathbb{OP}^2 \# \Sigma$ and $\mathbb{OP}^2$ which we can destabilize as once again the kernel module of $\pi_8(\mathbb{OP}^2)$ dies.

In [12] it is shown that in both dimensions 8 and 16 there are in fact countably many almost-closed manifolds (i.e. closed manifolds minus a disc) with $H_* \cong \mathbb{Z}^3$ determined by p_2 and p_4 of the tangent bundles respectively. The authors in this paper in fact use this to show that such closed manifolds are up to diffeomorphism uniquely determined by p_2^2 and p_4^2 respectively.

6 Conclusion

This concludes our work on developing some of the techniques used in classifying high-dimensional manifolds. We developed some of the theory of handlebody decompositions and surgery theory and used it to prove the h -cobordism theorem and from there Kreck's stabilization and destabilization theorems. Our two examples $\mathbb{HP}^2$ and $\mathbb{OP}^2$ at the end are easy special cases of the work in [12] of classifying manifolds M with $H_*(M) \cong \mathbb{Z} \oplus \mathbb{Z} \oplus \mathbb{Z}$. In fact the authors in that work use a refinement of the surgery methods of Kreck described in this essay to obtain a complete classification of such manifolds in terms of the Pontryagin class $p(TM)$.

Almost everything in this essay can be put into a workable framework in the non simply-connected case. The main obstruction we have avoided is Whitehead Torsion, which essentially measures when a

homotopy equivalence is a so-called *simple* homotopy equivalence (i.e. one induced by certain simple operations on a cell-structure). The analogue of the h -cobordism theorem is the s -cobordism theorem, where the inclusions $\partial_{0,1}W \hookrightarrow W$ are required to be simple homotopy equivalences. The cobordism is trivial in this case (See [13]). Whitehead torsion always vanishes when π_1 is trivial. The statements of the stabilization and destabilization theorems are the same with minor modifications in this case but the proof of destabilization is quite a lot more involved (one needs to control surgery-obstruction groups of matrices which take elements in $\mathbb{Z}[\pi]$. This can make things much more involved than the $\mathbb{Z}$ case, for instance via torsion).

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